Choosing and Combining Factoring Techniques
Answer Key
Algebra 1

Quick Answers

1. $3x\,(2x+3)$
2. $(x+3)\,(x+4)$
3. $(x-8)\,(x+8)$
4. $(x-5)\,(x+2)$
5. $(2x+1)\,(x+3)$
6. $4\,(x-3)\,(x+3)$
7. $(7x-5)\,(7x+5)$
8. $(3x-4)\,(x-2)$
9. $(x-3)\,(x-8)$
10. $2x^2+3x-20$
11. $2x\,(x-6)\,(x+2)$
12. (a) $=(x-2)\,(x+2)\,(x-3)\,(x+3)$, —

What is verified

11 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
HSA-APR.A.1 – *problem 10*
HSA-SSE.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12*
Difficulty 1–5 of 5 across 13 tagged checks.

1. Factor completely: $6x^2+9x$

Solution: Question 1 — is there a common factor? Yes: $6x^2$ and $9x$ share $3x$. Two terms remain inside, with no minus sign between squares, so nothing further factors.

$$\boxed{3x(2x+3)}$$

2. Factor completely: $x^2+7x+12$

Solution: No common factor; three terms with a leading coefficient of 1. Look for two numbers whose product is 12 and whose sum is 7: those are 3 and 4. $\boxed{(x+3)(x+4)}$

3. Factor completely: x^2-64

Solution: No common factor; two terms, both perfect squares, separated by a minus sign. That is a^2-b^2 with $a=x$ and $b=8$. $\boxed{(x-8)(x+8)}$

4. Factor completely: $x^2-3x-10$

Solution: Three terms, leading coefficient 1. Product -10, sum -3. Because the product is negative the two numbers have opposite signs: -5 and $+2$. $\boxed{(x-5)(x+2)}$

5. Factor completely: $2x^2+7x+3$

Solution: No common factor (2, 7 and 3 share nothing), and the leading coefficient is not 1, so use grouping. Find two numbers with product $2 \cdot 3 = 6$ and sum 7: those are 6 and 1.

$$\begin{aligned} 2x^2+7x+3 &= 2x^2+6x+x+3 \\ &= 2x(x+3)+1(x+3) \\ &= (2x+1)(x+3) \end{aligned}$$

$\boxed{(2x+1)(x+3)}$

6. Factor completely: $4x^2-36$

Solution: Common factor first: every term is divisible by 4. Only then is the leftover a difference of squares.

$$\begin{aligned} 4x^2-36 &= 4(x^2-9) \\ &= 4(x-3)(x+3) \end{aligned}$$

Stopping at $(2x-6)(2x+6)$ is not completely factored — each binomial still holds a factor of 2. $\boxed{4(x-3)(x+3)}$

7. Factor completely: $49x^2-25$

Solution: No common factor. Two terms, and each is a perfect square: $49x^2=(7x)^2$, $25=5^2$. $\boxed{(7x-5)(7x+5)}$

8. Factor completely: $3x^2-10x+8$

Solution: No common factor, leading coefficient 3, so use grouping. Product $3 \cdot 8 = 24$, sum -10; both numbers are negative: -6 and -4.

$$\begin{aligned} 3x^2-10x+8 &= 3x^2-6x-4x+8 \\ &= 3x(x-2)-4(x-2) \\ &= (3x-4)(x-2) \end{aligned}$$

$\boxed{(3x-4)(x-2)}$

9. Factor completely: $x^2-11x+24$

Solution: Leading coefficient 1. Product 24, sum -11: both factors negative, -3 and -8. $\boxed{(x-3)(x-8)}$

10. A classmate says that $2x^2+3x-20$ factors as $(2x-5)(x+4)$. Multiply the two binomials out and write the product in standard form, so that the claim can be compared with the original expression.

Solution: Multiply every term of the first binomial by every term of the second, then combine the two middle products.

$$\begin{aligned} (2x-5)(x+4) &= 2x^2+8x-5x-20 \\ &= 2x^2+3x-20 \end{aligned}$$

The product matches the original expression, so the classmate is right. $\boxed{2x^2 + 3x - 20}$

11. Factor completely: $2x^3 - 8x^2 - 24x$

Solution: Every term contains $2x$; take it out first. What is left is a monic trinomial with product -12 and sum -4, giving -6 and $+2$.

$$\begin{aligned} 2x^3 - 8x^2 - 24x &= 2x(x^2 - 4x - 12) \\ &= 2x(x-6)(x+2) \end{aligned}$$

$$\boxed{2x(x-6)(x+2)}$$

12. **Synthesis challenge.** $x^4 - 13x^2 + 36$. (a) Factor completely. (b) Explain which technique you used first and why that technique was available, then name the technique that finished the job.

Solution (a): There is no common factor, and there are three terms — so this behaves like a trinomial, with x^2 playing the part of the variable (the exponents 4 and 2 are double 2 and 1). Product 36, sum -13, so the numbers are -4 and -9:

$$\begin{aligned} x^4 - 13x^2 + 36 &= (x^2 - 4)(x^2 - 9) \\ &= (x-2)(x+2)(x-3)(x+3) \end{aligned}$$

Each quadratic factor is itself a difference of two squares, so the work is not finished at line one. $\boxed{(x-2)(x+2)(x-3)(x+3)}$

(b) — instructor-judged. A full answer names trinomial factoring as the first move and says why it applies: three terms, no common factor, and the exponents 4 and 2 are double 2 and 1, so x^2 behaves like a single variable. It then names the difference of two squares as the technique that finishes the job, applied twice. Naming only one of the two techniques, or calling the original expression a difference of squares, is partial credit at best.